Project Plan

Minor AI for Society

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1. Document history

Table 1. Document History

Version	Date	Status	Author	Description	Remarks
0.1	11-03-2025	First Version	Casey Stijlaart	Project Assignment	-

2. Project Assignment

2.1. Context

This project is part of the AI for Society minor at Fontys Eindhoven, which focuses on understanding and applying AI to practical problems that can benefit individuals and society. Indoor plants are a common feature in homes and offices, but many people struggle to provide the right care, leading to poor plant health and reduced benefits such as improved air quality and wellbeing.

By developing a smart desk plant monitoring system, this project applies AI techniques to a real-world context, providing automated insights and care recommendations for plants. The project also serves as a personal learning opportunity to gain hands-on experience with AI and embedded software, bridging theoretical knowledge from the minor with practical application.

2.2. Project Goal

The goal of this project is to gain practical experience with integrating AI into a practical system. To support this goal, a smart desk plant monitoring system will be developed that provides real-time insights and care recommendations for indoor plants.

2.3. Assignment

In this personal project, the assignment is to design and implement a system that makes use of AI. The system will be a smart desk plant monitoring system that provides real-time insights and care recommendations for indoor plants. The primary objective of this project is to develop practical experience and expand knowledge in AI and embedded software development.

2.4. Finished Product

The finished product of this project will be a fully functional smart desk plant monitoring system that combines AI software and embedded hardware.

Key features of the system include:

- Real-time monitoring of plant health using sensors (e.g., soil moisture, light, humidity).
- AI-driven analysis of sensor data to provide care recommendations for each plant.
- User-friendly interface for displaying plant status and actionable advice.
- Documentation of system architecture, AI models, and implementation details to support future improvements or adaptation.

The final deliverable will be a working prototype that demonstrates the system’s capabilities in a practical desk environment, along with supporting documentation that explains the design, AI integration, and ethical and legal considerations addressed during development.

2.5. Research Questions

To successfully deliver the project, several key aspects must be explored. Each of these aspects plays an important role in determining the functioning and usability of the system. To investigate these areas thoroughly, research questions have been formulated, each addressing a specific focus.

2.5.1. Main Question

How can a smart desk plant monitoring system use AI to provide accurate, user-friendly care recommendations while addressing ethical, legal, and societal considerations?

2.5.2. Sub-questions

Technical / AI-related

- Which type of AI is most suitable for monitoring plant health and generating care recommendations?
- Are there existing models that can be used or adapted for this purpose, or is it necessary to develop a custom model?
- Which sensors and data collection methods provide the most reliable and relevant input for real-time plant monitoring?
- How can the system be implemented effectively in a small embedded setup?

User / Human Interaction

- How can the system present care recommendations in a clear and actionable way to users?
- Which features improve usability and engagement for desk plant monitoring?

Ethical and Legal Considerations

- How can user data be collected, stored, and processed while maintaining privacy and security?
- What ethical considerations arise from automating plant care, and how should they influence system design?

2.6. Research Methods

In this project, the DOT-framework is used as primary research methodology. The DOT framework is a systematic approach that focusses on various research strategies to build system. The research strategies are Field, Lab, Library, Workshop, and Showroom research.

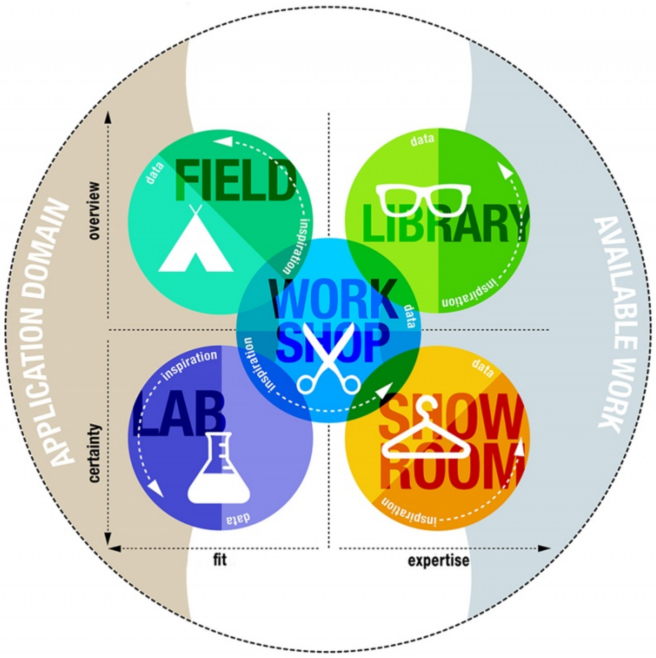

Figure 1. DOT framework

2.6.1. Research Methods x Research Questions

Each research question is met with a corresponding research strategy and technique based on the DOT framework. This approach ensures that every research question is answered effectively.

For a comprehensive view of how the DOT framework aligns with our project, consult [Research Methods and Research Questions].

Table 2. Research Methods and Research Questions

Research Question	**Strategy**	**Technique**	**Steps**
Which type of AI is most suitable for monitoring plant health and generating care recommendations?	Literature review & prototyping	Compare AI models (rule-based, ML, neural networks)	1. Review existing AI approaches for plant monitoring 2. Select candidate models 3. Prototype and evaluate performance
Are there existing models that can be used or adapted, or is a custom model needed?	Literature review & evaluation	Analyze pre-trained models	1. Search for existing plant health datasets and models 2. Evaluate suitability for smart desk setup 3. Decide whether to adapt or build custom model
Which sensors and data collection methods provide the most reliable input for real-time monitoring?	Experimentation & prototyping	Sensor testing	1. Select candidate sensors (soil moisture, light, humidity, etc.) 2. Collect sample data 3. Analyze reliability and accuracy
How can the system be implemented effectively in a small embedded setup?	Prototyping & testing	Embedded system development	1. Design embedded architecture 2. Implement system with selected sensors and AI 3. Test real-time performance
How can the system present care recommendations clearly to users?	User-centered design	Mockups & user testing	1. Design interface mockups 2. Conduct small user tests 3. Collect feedback and iterate
Which features improve usability and engagement for desk plant monitoring?	Observation & feedback	User surveys & interviews	1. Implement prototype with basic features 2. Observe users interacting with system 3. Collect feedback on usability and engagement

Research Question	Strategy	Technique	Steps
How can user data be collected, stored, and processed while maintaining privacy and security?	Literature & benchmarking	Review legal/ethical frameworks	1. Study GDPR and privacy requirements 2. Analyze best practices for IoT data handling 3. Design compliant data handling processes
What ethical considerations arise from automating plant care, and how should they influence system design?	Literature & reflection	Ethical assessment	1. Identify potential ethical concerns (automation, over-reliance) 2. Reflect on implications for design 3. Incorporate safeguards in system design

3. Approach and Planning

3.1. Approach

Throughout the project, I am following agile principles, which allow for flexibility and adaptability as the project progresses. This approach enables me to iteratively develop and refine the system based on feedback and testing results, ensuring that the final product meets the desired goals and requirements.

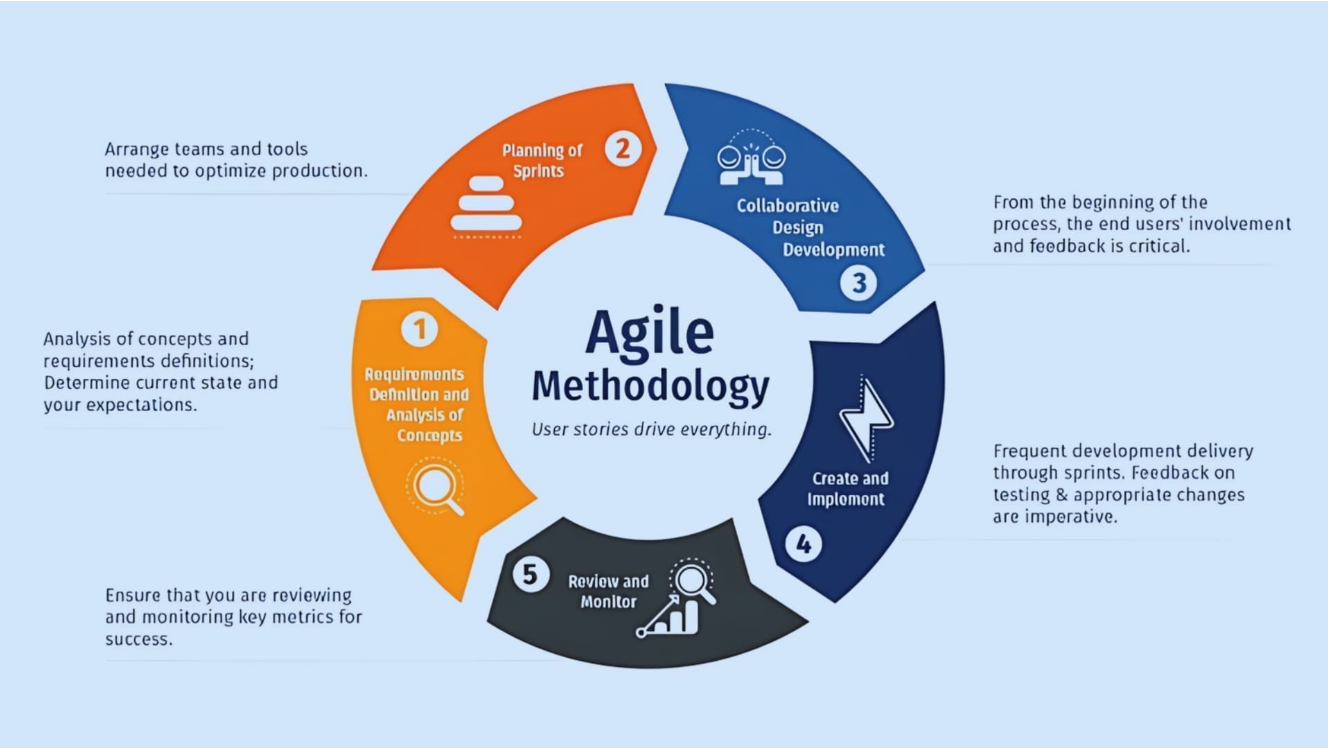

Figure 2. Agile Workflow

3.1.1. Test Approach

To ensure a seamless software implementation, I have integrated testing into the project. Testing takes different forms, each focusing on different aspects of the project. The primary testing methods employed in this project are unit testing and integration testing, which help ensure the quality and functionality of the software.

3.2. Breakdown Project

The project follows an Agile workflow, enabling iterative development and continuous improvement. The main phases are:

1. **Requirements & Concept** : Define project goals, functional and non-functional requirements, and create an initial backlog of tasks.
2. **System Design** : Plan the overall architecture, including hardware-software interactions, modules, and interfaces.
3. **Sprint Planning** : Select tasks from the backlog for each sprint and break them into actionable

development items.

4. **Implementation** : Develop firmware and software modules incrementally, integrating with hardware.
5. **Testing & Integration** : Perform unit tests, integration tests, and hardware-in-the-loop testing to ensure correctness and reliability.
6. **Review & Retrospective** : Demonstrate progress, gather feedback, and reflect on improvements for the next sprint.

This iterative approach ensures flexibility, early detection of issues, possibilties for adaptions based upon feedback and continuous delivery of working system increments.

3.3. Timeline

The project's overall timeline, depicted in [Project Timeline], offers a high-level overview of project phases and significant events.

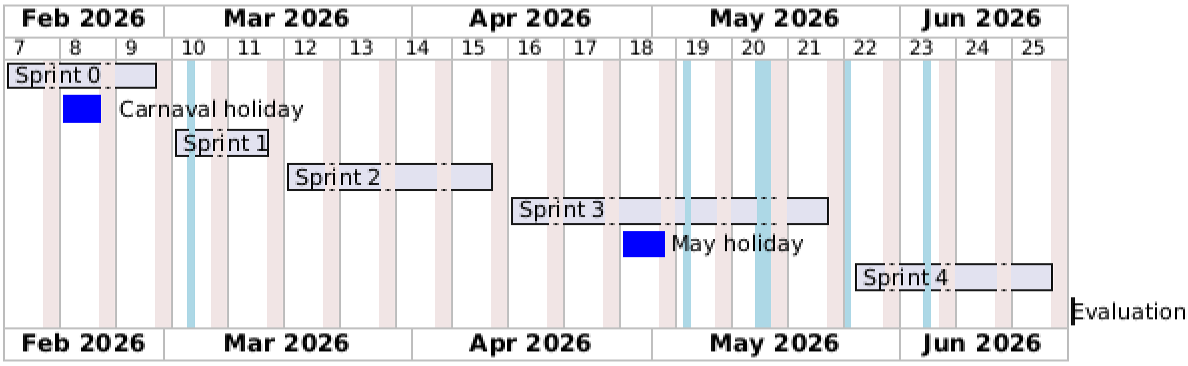

Figure 3. Project Timeline